

Don't Shit where You Drill!



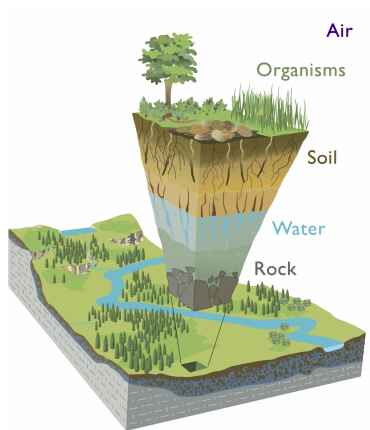
Don't Shit where You Drill!

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**MELLEMFOLKELIGT
SAMVIRKE **

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¹ize Illustration of soil layers in the near surface [Chorover et al., 2007].

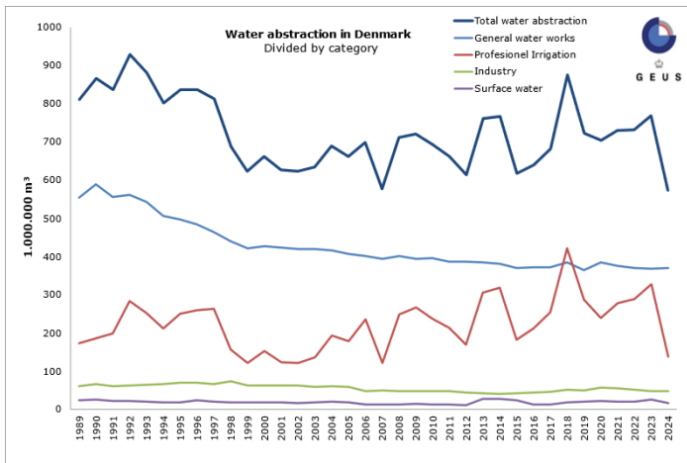
¹_s

ToC

Part One:

- ▶ How are we looking?
- ▶ What are the culprits?
- ▶ What is the government doing? Promising?

DENMARK'S WATER



Data collected by GEUS 2025 [Lærke Thorling et al., 2025].

Nitrate - the Classic

CONSEQUENCES

Environmental Pollution of bodies of water. Excess algae growth, extinction of fauna etc.

CONSEQUENCES FROM THE TAB

“Nitrate in drinking water is a contaminant which can affect human health and has been associated with an increased risk of, amongst other diseases, colorectal cancer”. [Jacobsen et al., 2024]

For over 60 years, the global guideline for nitrates in drinking water has been 50 milligrams per litre (mg/L) [...] set in the 1950s².

²<https://www.greenpeace.org/international/story/82511/industrial-agriculture-profits-toxic-nitrates-pollution-tap-water/>

This limit is hopelessly out of date

Literature proposes lower limits of $< 10\%$ reducing the risk by 15%. Even lower limits of $< 4\%$ bring this number down further [Jacobsen et al., 2024, Schullehner et al., 2018].

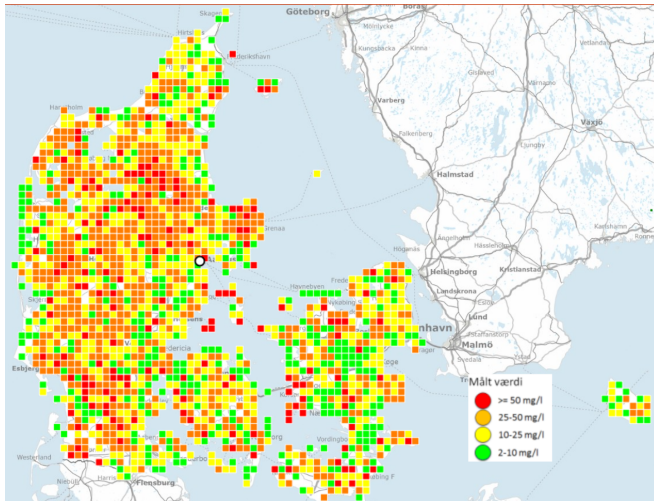
BLUE BABIES (NOT GOOD)



Infant methemoglobinemia

- ▶ Nitrate prevents oxygen from binding to their blood. Can progress to cause coma and death if not left untreated [Knobeloch et al., 2000, Chen and Nappe, 2026].
- ▶ Water used in formula contained nitrate-nitrogen concentrations of 22.9 and 27.4 mg/L [Knobeloch et al., 2000].

NITRATE POLLUTION IN DENMARK



<https://www.geus.dk/produkter-ydelser-og-faciliteter/data-og-kort/grundvandskort-og-data>

MIXING AND EXPLOTING

PFAS - the Immortal

WHERE DO THEY LIVE?

Where **DON'T** they live!? PFAS are found in water, air, fish, and soil..

³<https://www.epa.gov/pfas/pfas-explained>

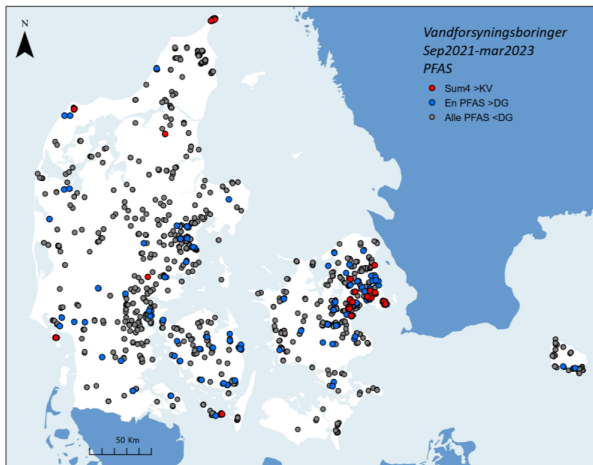
WHERE DO THEY LIVE?

Where DON'T they live!? PFAS are found in water, air, fish, and soil..

Also, in blood of people, animals, in food products³

³<https://www.epa.gov/pfas/pfas-explained>

PFAS MEASUREMENTS IN GROUNDWATER WELLS



Red: PFAS chemicals above limit. *Blue:* Measured PFAS chemicals. *Grey:* None detected [Anders R. Johnsen, Lærke Thorling & Christian N. Albers, 2023].

ToC

Part Two:

- ▶ Case studies in the global south
- ▶ How are we as Europeans involved?
- ▶ How is climate change changing the picture?

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